MILITARY SPECIFICATION

INDICATOR, TACHOMETER, HELICOPTER, 0-4000 RPM,
TYPE MA-1

This amendment forms a part of Military Specification MIL-I-8580B dated 2 April 1965, and is approved for use by all Departments and Agencies of the Department of Defense.

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Paragraph 2.1 Under Military (continued) - Delete reference to:

MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-R-22256	Reliability Requirements for Design of Electronic Equipment or Systems
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists

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Paragraph 2.1 Under STANDARDS Military - Add:

MIL-STD-100	Engineering Drawing Practices
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

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Paragraph 3.8.1 Special reliability design - Delete.

FSC 6620

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3.8.2 Reliability in mean-time-between-failures (MTBF) - Delete the body of this paragraph and substitute the following:

"The indicator shall have a specified MTBF (θ_o) of 1500 hours between failures when tested and accepted as outlined under the requirements of 4.4.3."

Paragraph 3.10 Interchangeability - Delete, and substitute the following:

"3.10 Part numbering of interchangeable parts - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto."

Paragraph 3.13 - Delete this paragraph.

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Paragraph 4.4.2 Sampling plans - In the third sentence, immediately after the word "refurbished" insert:

", if necessary,"

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Paragraph 4.4.3 Reliability tests - Delete, and substitute the following paragraphs:

"4.4.3 Reliability assurance tests - Reliability Assurance Tests shall be conducted in accordance with MIL-STD-781. Qualification Phase and Production Acceptance (Sampling) Phase Tests shall be conducted. Equipments selected for Reliability Assurance Tests shall first have passed the Individual Tests.

4.4.3.1 Reliability qualification phase - A sample of at least three, but not more than six, preproduction equipments shall be tested as outlined in MIL-STD-781, under the section entitled 'Qualification Phase of Production Reliability Tests'. An alternate method of performing this phase of testing is selection of sample units from the first production lot; however, the lot may not be delivered until an accept decision is reached. If the alternate method is chosen, the number of test samples shall be in accordance with Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

4.4.3.2 Reliability production acceptance (sampling) phase tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled 'Production Acceptance (Sampling) Phase of Production

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Reliability Tests'. For the Sampling Phase,Test Level E shall be used. The Accept-Reject Criteria for Test Plan II shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Procedure for production acceptance (sampling) phase - The Reliability Production Acceptance (Sampling) Phase tests shall be conducted at least once for each lot. The number of equipments shall be in accordance with Table 5 of MIL-STD-781. The test program shall start with the first month's production unless that lot is used for the Qualification Phase, in which case Acceptance (Sampling) Phase shall begin with the second month's production. Equipments shall be selected and placed on test in the same manner as though one month's production was a lot. The equipments shall be tested until an accept or reject decision is reached. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached pending a review of the contractor's efforts to improve the equipment, the equipment quality control. etc.

4.4.3.3 Procedures report and test details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, etc., shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The following paragraphs shall be considered as minimum requirements and apply to both phases.

4.4.3.3.1 Duty cycle - The duty cycle shall be continuous. The duty cycle shall consist of operating the indicator as described in the Endurance test, i.e., 2600 $\pm$200 rpm.

4.4.3.3.2 Performance characteristics to be measured - The Scale error at room temperature and friction error tests, described under Inspection methods, shall be conducted at least once each week. For equipments with less than 500 operating hours those test tolerances shall apply; for equipments with greater than 500 operating hours an additional 5 rpm shall be allowed. The indicator shall be observed daily, on regular working days, for any obvious failure such as rough pointer movement.

4.4.3.3.3 Failure criteria - Whenever performance characteristics fall below the acceptance requirements (4.4.3.3.2), at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one part caused the other parts to fail.

4.4.3.3.4 Preventive maintenance - No preventive maintenance may be accomplished on the equipments while they are on test.

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4.4.3.3.5 Operational stability - No adjustment of any controls inaccessible to the operator shall be made during the Reliability Test.

4.4.3.3.6 Disposition of samples upon completion of tests - Any equipment tested may be delivered on contract provided it meets all the following requirements:

a. It is representative of production units currently being accepted.

b. It is in "good as new" condition or has been refurbished (see 6.4.3 and 6.4.4).

c. It is otherwise satisfactory."

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Paragraph 6.2 Ordering data - Delete item (f).

Paragraph 6.3 Data - Delete this paragraph and subparagraphs. Add the following new paragraph:

"6.2.1 Contract data requirements - Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.4	Test reports	
4.4.3.2.1	Reliability test results	
4.4.3.3	Reliability procedures and test details	

Such data will be delivered as described on approved (numbered) DID's (Data Item Description/DD Form 1664) when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into the applicable contract."

Paragraph 5. PREPARATION FOR DELIVERY - Substitute MIL-STD-794 for MIL-P-7936 throughout this Section.

Custodians:
Army - AV
Navy - AS

Preparing Activity:
Navy - AS
Project No. 6620-0271

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